
saad@sand-dweller: ~/xv6-labs-2026 — make qemu
~/xv6-labs-2026
sepc=0x49be stval=0x32013000
usertrap(): unexpected scause 0xf pid=6631
sepc=0x49be stval=0x33013000
usertrap(): unexpected scause 0xf pid=6632
sepc=0x49be stval=0x34013000
usertrap(): unexpected scause 0xf pid=6633
sepc=0x49be stval=0x35013000
usertrap(): unexpected scause 0xf pid=6634
sepc=0x49be stval=0x36013000
usertrap(): unexpected scause 0xf pid=6635
sepc=0x49be stval=0x37013000
usertrap(): unexpected scause 0xf pid=6636
sepc=0x49be stval=0x38013000
usertrap(): unexpected scause 0xf pid=6637
sepc=0x49be stval=0x39013000
usertrap(): unexpected scause 0xf pid=6638
sepc=0x49be stval=0x3a013000
usertrap(): unexpected scause 0xf pid=6639
sepc=0x49be stval=0x3b013000
usertrap(): unexpected scause 0xf pid=6640
sepc=0x49be stval=0x3c013000
usertrap(): unexpected scause 0xf pid=6641
sepc=0x49be stval=0x3d013000
usertrap(): unexpected scause 0xf pid=6642
sepc=0x49be stval=0x3e013000
usertrap(): unexpected scause 0xf pid=6643
sepc=0x49be stval=0x3f013000
OK
test lazy_copy: OK
test lazy_copyinstr: OK
test lazy_sbrk: OK
test partial_write: OK
test unlinkcwd: OK
ALL TESTS PASSED
$


saad@sand-dweller: ~/xv6-labs-2026 — make qemu
~/xv6-labs-2026
mkfs/mkfs fs.img README user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_alarmtest user/_call user/_bttest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1088 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ alarmtest
test0 start
.............................alarm!
test0 passed
test1 start
..alarm!
..alarm!
..alarm!
..alarm!
..alarm!
..alarm!
..alarm!
..alarm!
..alarm!
.alarm!
test1 passed
test2 start
.....................alarm!
test2 passed
test3 start
test3 passed
$

saad@sand-dweller: ~/xv6-labs-2026 — riscv64-unknown-elf-addr2line -e kernel/kernel

~/xv6-labs-2026

```
  binutils-riscv64-unknown-elf

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 88
  Download size: 3,189 kB
  Space needed: 16.7 MB / 65.3 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/universe amd64 binutils-riscv64-unknown-elf amd64 2.45.50.20251209-1ubuntu1+7build1 [3,189 kB]
Fetched 3,189 kB in 3s (963 kB/s)
Selecting previously unselected package binutils-riscv64-unknown-elf.
(Reading database… 215204 files and directories currently installed.)
Preparing to unpack …/binutils-riscv64-unknown-elf_2.45.50.20251209-1ubuntu1+7build1_amd64.deb…
Unpacking binutils-riscv64-unknown-elf (2.45.50.20251209-1ubuntu1+7build1)…
Setting up binutils-riscv64-unknown-elf (2.45.50.20251209-1ubuntu1+7build1)…
Processing triggers for man-db (2.13.1-1build1)…
saad@sand-dweller:~/xv6-labs-2026$ riscv64-unknown-elf-addr2line -e kernel/kernel
0x0000000080001f80
0x0000000080001e5a
0x0000000080001b76./kernel/sysproc.c:74
./kernel/syscall.c:150 (discriminator 1)
0x0000000080001f80
??:0
0x0000000080001e5a
./kernel/syscall.c:150 (discriminator 1)
0x0000000080001b76
./kernel/trap.c:81

??:0

??:0
0x0000000080001f80
./kernel/sysproc.c:74
0x0000000080001f80
./kernel/sysproc.c:74
```


saad@sand-dweller: ~/xv6-labs-2026 — make qemu
~/xv6-labs-2026
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_TRAPS -DLAB_TRAPS -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/call.o user/call.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_call user/call.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_call > user/call.asm
riscv64-linux-gnu-objdump -t user/_call | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/call.sym
riscv64-linux-gnu-gcc -Wall -Werror -Wno-unknown-attributes -O -fno-omit-frame-pointer -ggdb -gdwarf-2 -ffile-prefix-map=/home/saad/xv6-labs-2026=. -DSOL_TRAPS -DLAB_TRAPS -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen -fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fno-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie -no-pie -c -o user/bttest.o user/bttest.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T user/user.ld -o user/_bttest user/bttest.o user/ulib.o user/usys.o user/printf.o user/umalloc.o
riscv64-linux-gnu-objdump -S user/_bttest > user/bttest.asm
riscv64-linux-gnu-objdump -t user/_bttest | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > user/bttest.sym
mkfs/mkfs fs.img README user/_cat user/_echo user/_forktest user/_grep user/_init user/_kill user/_ln user/_ls user/_mkdir user/_rm user/_sh user/_stressfs user/_usertests user/_grind user/_wc user/_zombie user/_logstress user/_forphan user/_dorphan user/_sync user/_alarmtest user/_call user/_bttest
nmeta 47 (boot, super, log blocks 31, inode blocks 13, bitmap blocks 1) blocks 1953 total 2000
balloc: first 1102 blocks have been allocated
balloc: write bitmap block at sector 46
qemu-system-riscv64 -machine virt -bios none -kernel kernel/kernel -m 128M -smp 3 -nographic -global virtio-mmio.force-legacy=false -drive file=fs.img,if=none,format=raw,id=x0 -device virtio-blk-device,drive=x0,bus=virtio-mmio-bus.0
xv6 kernel is booting
hart 2 starting
hart 1 starting
init: starting sh
$ bttest
backtrace:
0x0000000080001f80
0x0000000080001e5a
0x0000000080001b76
$

```
S -DLAB_TRAPS -march=rv64gc -std=gnu99 -MD -mcmodel=medany -ffreestanding -fno-common -nostdlib -fno-builtin-strncpy -fno-builtin-strncmp -fno-builtin-strlen
-fno-builtin-memset -fno-builtin-memmove -fno-builtin-memcmp -fno-builtin-log -fno-builtin-bzero -fno-builtin-strchr -fno-builtin-exit -fno-builtin-malloc -fn
o-builtin-putc -fno-builtin-free -fno-builtin-memcpy -Wno-main -fno-builtin-printf -fno-builtin-fprintf -fno-builtin-vprintf -I. -fno-stack-protector -fno-pie
 -no-pie  -c -o kernel/spinlock.o kernel/spinlock.c
riscv64-linux-gnu-ld -z max-page-size=4096 -T kernel/kernel.ld -o kernel/kernel kernel/entry.o kernel/kalloc.o kernel/string.o kernel/main.o kernel/vm.o kerne
l/proc.o kernel/swtch.o kernel/trampoline.o kernel/trap.o kernel/syscall.o kernel/sysproc.o kernel/bio.o kernel/fs.o kernel/log.o kernel/sleeplock.o kernel/fi
le.o kernel/pipe.o kernel/exec.o kernel/sysfile.o kernel/kernelvec.o kernel/plic.o kernel/virtio_disk.o kernel/start.o kernel/console.o kernel/printk.o kernel
/uart.o kernel/spinlock.o
/usr/bin/riscv64-linux-gnu-ld.bfd: warning: kernel/kernel has a LOAD segment with RWX permissions
riscv64-linux-gnu-objdump -S kernel/kernel > kernel/kernel.asm
riscv64-linux-gnu-objdump -t kernel/kernel | sed '1,/SYMBOL TABLE/d; s/ .* / /; /^$/d' > kernel/kernel.sym
make[1]: Leaving directory '/home/saad/xv6-labs-2026'
== Test answers-traps.txt ==
answers-traps.txt: OK
== Test backtrace test ==
$ make qemu-gdb
backtrace test: OK (6.0s)
== Test running alarmtest ==
$ make qemu-gdb
(4.4s)
== Test   alarmtest: test0 ==
  alarmtest: test0: OK
== Test   alarmtest: test1 ==
  alarmtest: test1: OK
== Test   alarmtest: test2 ==
  alarmtest: test2: OK
== Test   alarmtest: test3 ==
  alarmtest: test3: OK
== Test usertests ==
$ make qemu-gdb
usertests: OK (105.0s)
== Test time ==
time: OK
Score: 95/95
saad@sand-dweller:~/xv6-labs-2026$
```